

DETAILED DESCRIPTION OF THE INVENTION

Alternative Embodiments and Variations

The systems and methods described herein may be implemented in a variety of structural, electronic, and computational configurations without departing from the core inventive concepts of synchronized radial distance acquisition and longitudinal motion tracking for penile anatomical geometry reconstruction.

In certain embodiments, the housing structure may comprise a handheld wand, a partially enclosing C-shaped frame, an adjustable band, a sliding rail mechanism, or a semi-rigid guide assembly configured to constrain motion along a defined pathway. The device may further include removable or interchangeable sensor modules to allow adaptation to varying resolution requirements or hardware upgrades.

Time-of-flight sensing may be implemented using single-point emitters, multi-zone depth arrays, structured light projectors, or other optical depth measurement technologies capable of resolving radial distance to an anatomical surface. Sensor placement may vary along the longitudinal axis, around circumferential regions, or at angular offsets to optimize contour sampling. In some embodiments, depth sensing may be augmented by additional proximity sensors or capacitive detection modules.

Optical motion sensing may be implemented using one or more optical flow sensors, surface-relative tracking cameras, emitter-receiver displacement pairs, or other motion detection architectures capable of resolving linear displacement relative to tissue. Motion tracking may alternatively be inferred through mechanical encoders, magnetic tracking systems, or hybrid sensing approaches combining optical and inertial inputs.

Processing architecture may be fully integrated within the device housing, partially distributed between the device and a companion application, or entirely executed in a remote computing environment. Data reconstruction may utilize deterministic mathematical models, machine learning regression models, neural network-based surface prediction, or adaptive hybrid modeling techniques.

In certain embodiments, the system may perform segmented scanning in which multiple discrete regions are scanned independently and subsequently merged into a unified geometry model. Bidirectional scanning paths, rotational sampling techniques, or adaptive sampling density adjustments may be implemented to increase reconstruction fidelity.

The inventive concepts disclosed herein are not limited to any specific combination of sensor count, housing geometry, computational method, communication protocol, or deployment configuration. Any embodiment capable of acquiring synchronized distance and positional data sufficient to generate a user-specific penile anatomical geometry profile falls within the scope of the present disclosure.